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Case 33-2022: An 11-Year-Old Girl with Redness of the Eyes

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PRESENTATION OF CASE

Dr. Natalie A. Diacovo (Pediatrics): An 11-year-old girl was evaluated in the rheumatology clinic of this hospital because of redness of the eyes.

The patient had been well until 8 weeks before the current evaluation, when she noticed mild eye redness, which was worse in the right eye than in the left eye. There was intermittent swelling of the eyelids but no photophobia, itching, pain, tearing, or discharge. The patient's parents administered naphazoline hydrochloride and glycerin eye drops, but the eye redness did not abate. When the eye redness had persisted for 2 weeks, the patient was taken to the pediatrics clinic of another hospital. She was given a 7-day course of an oral antibiotic agent as empirical treatment for possible preseptal cellulitis.

Five weeks before the current evaluation, the eye redness persisted despite treatment, and the patient was referred to the ophthalmology clinic of the other hospital. The blood pressure was not measured during this evaluation. The visual acuity with correction was 20/20 in the right eye and 20/25 in the left eye. The pupils were symmetric and reactive to light, with no relative afferent pupillary defect. Slit-lamp examination revealed abundant white cells in the anterior chamber of both eyes, with a greater amount in the right eye than in the left eye. There were also keratic precipitates, posterior synechiae, and rare iris nodules in the right eye. Funduscopic examination was notable for optic-disk swelling and peripheral retinal hemorrhages in both eyes, as well as perivascular exudates in the right eye. Cyclopentolate eye drops in the right eye and prednisolone eye drops in both eyes were prescribed.

During the subsequent 3 weeks, additional tests were performed in the pediatrics clinic of the other hospital. Screening blood tests for human immunodeficiency virus (HIV) types 1 and 2, syphilis, and Lyme disease were negative. An interferon-γ release assay for *Mycobacterium tuberculosis* was indeterminate. Other laboratory test results are shown in Table 1. Radiography of the chest revealed hilar fullness.

Dr. Maria G. Figueiro Longo: Computed tomography (CT) of the chest and abdomen (Fig. 1A and 1B), performed after the administration of intravenous contrast material, showed a normal-appearing thymus. However, diffuse lymphadenopathy was

Table 1. Laboratory Data.

Variable	Reference Range, Other Hospital	3–6 Wk after Symptom Onset, Pediatrics Clinic, Other Hospital	6 Wk after Symptom Onset, Emergency Department, Other Hospital
Hemoglobin (g/dl)	11.3–13.4	—	10.2
Hematocrit (%)	32.3–38.3	34.0	31.3
Platelet count (per μl)	187,000–376,000	175,000	233,000
White-cell count (per μl)	5410–9700	5600	4690
Differential count (per μl)			
Neutrophils	2580–5950	—	2960
Lymphocytes	1230–2760	—	900
Monocytes	190–810	—	610
Eosinophils	40–190	—	160
Basophils	10–50	—	20
Creatinine (mg/dl)*	0.30–1.00	—	0.75
Alanine aminotransferase (U/liter)	3–30	28	17
Aspartate aminotransferase (U/liter)	2–40	34	31
Alkaline phosphatase (U/liter)	60–335	—	129
Albumin (g/dl)	3.0–4.6	—	4.2
Total protein (g/dl)	5.5–8.2	—	8.8
C-reactive protein (mg/liter)	0–5.0	2.8	3.6
Erythrocyte sedimentation rate (mm/hr)	0–30	—	90
Angiotensin-converting enzyme (U/liter)	24–121	74	—
Lysozyme (μg/ml)	0–2.75	2.60	3.83
Antinuclear antibody	Negative	Negative	Negative
Antineutrophil cytoplasmic antibody	Negative	Negative	Negative

* To convert the values for creatinine to micromoles per liter, multiply by 88.4.

detected in the mediastinum, upper abdomen, and axillae, with the largest lymph node measuring 2.0 cm by 1.3 cm by 1.8 cm.

Dr. Diacovo: Two weeks before the current evaluation, the patient was asked to present to the emergency department of the other hospital for an expedited workup. On evaluation, she described eye redness that had persisted but decreased in severity. She reported no fever, weight loss, night sweats, rash, headache, cough, shortness of breath, or joint pain. On examination, the temperature was 37.0°C, the blood pressure 96/61 mm Hg, the heart rate 95 beats per minute, the respiratory rate 20 breaths per minute, and the oxygen saturation 100% while she was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 18.3.

The patient appeared well, with no oral lesions or rash. The visual acuity was unchanged. Slit-lamp examination revealed keratic precipitates and white cells in the anterior chamber of both eyes, with a greater amount in the right eye than in the left eye. There were extensive posterior synechiae in the right eye. Funduscopic examination (Fig. 2) showed vitreous white cells, optic-disk swelling, perivascular exudates, peripheral retinal hemorrhages, and cotton-wool spots in both eyes. The lungs were clear on auscultation. Urinalysis was normal. Other laboratory test results are shown in Table 1. Methylprednisolone was administered, and the patient was admitted to the other hospital.

During the next 4 days, additional tests were performed, and sputum specimens were obtained for an acid-fast bacilli smear and culture. Optical coherence tomography revealed optic-disk swelling but no cystoid macular swelling. Fluorescein angiography revealed optic-disk leakage with areas of peripheral retinal nonperfusion and some vessel leakage.

Dr. Figueiro Longo: Magnetic resonance imaging (MRI) of the head (Fig. 1C), performed before and after the administration of intravenous con-

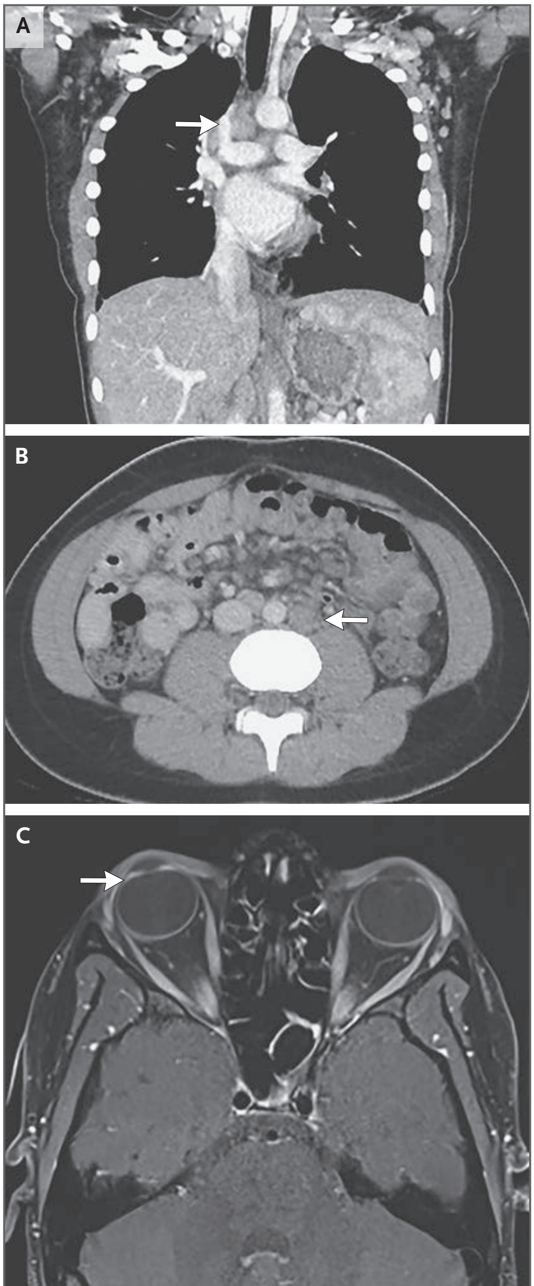

Figure 1. Initial Imaging Studies.

CT of the chest and abdomen was performed after the administration of intravenous contrast material. A coronal image of the chest (Panel A) shows mediastinal lymphadenopathy (arrow). An axial image of the abdomen (Panel B) shows retroperitoneal lymphadenopathy (arrow). MRI of the head was performed after the administration of intravenous contrast material. An axial T1-weighted fat-suppressed image of the orbits (Panel C) shows enhancement of the irises, with greater enhancement in the right globe (arrow) than in the left globe, a finding consistent with the clinical diagnosis of uveitis.

trast material, showed protrusion and enhancement of the optic disks and enhancement of the irises, a finding consistent with the clinical diagnosis of uveitis. There was no evidence of optic neuritis or perineuritis.

Dr. Diacovo: The administration of cyclopentolate eye drops in the right eye and prednisolone eye drops in both eyes was continued. Treatment with oral prednisone was started, and a regimen of rifampin, isoniazid, ethambutol, and pyrazinamide was begun as empirical treatment for possible tuberculosis. On the fifth hospital day, the patient was discharged home. She was instructed to follow up in the ophthalmology clinic of the other hospital and was referred to the rheumatology clinic of this hospital.

In the rheumatology clinic, additional history was obtained. The patient had had normal growth and development and was currently in middle school. She had received all routine childhood vaccinations. She had emigrated from West Africa to the United States 6 years earlier; since then, she had lived with her mother and father in a suburban town in New England and had not traveled outside the region. Six months before this evaluation, an interferon-γ release assay for *M. tuberculosis* had been positive. At that time, the patient had received a prescription for a medication to treat latent tuberculosis infection; however, the patient's parents could not recall giving the medication to the patient. She was taking the prescribed cyclopentolate and prednisolone eye drops, oral prednisone, and regimen of rifampin, isoniazid, ethambutol, and pyrazinamide. There were no known drug allergies. The patient's parents were healthy.

A diagnosis and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Ezra M. Cohen: This 11-year-old girl presented with redness of the eyes and diffuse lymphadenopathy identified on CT, in the absence of other reported systemic symptoms. Other clinically significant features of her presentation include a positive interferon-γ release assay 6 months earlier, an intermittently low platelet count and lymphopenia, an intermittently elevated lysozyme level, elevated levels of inflammatory markers, negative tests for antinuclear antibodies and antineutrophil cytoplasmic antibodies, and enhancement of the optic disks on MRI.

In challenging diagnostic cases such as this one, I find it helpful to focus on the most unique clinical manifestation of the patient's presentation when developing a differential diagnosis. In this patient, retinal vasculitis is the most unusual feature of her presentation, because it is a rare finding in children.

RETINAL VASCULITIS

Retinal vasculitis can be caused by inflammatory, infectious, and primary ocular conditions. This patient had elevated levels of systemic inflammatory markers and diffuse lymphadenopathy, findings that make an isolated primary ocular disorder less likely than an inflammatory or infectious cause. Inflammatory causes of retinal vasculitis to consider in this patient include sarcoidosis, granulomatosis with polyangiitis (GPA), Behçet's disease, Susac's syndrome, systemic lupus erythematosus, polyarteritis nodosa, HLA-B27–associated diseases, relapsing polychondritis, and the Vogt–Koyanagi–Harada disease. Infectious causes include tuberculosis, Lyme disease, cat scratch disease, toxoplasmosis, and herpesvirus infection.[1]

SARCOIDOSIS

From this list of potential diagnoses, sarcoidosis is the most likely diagnosis for a few reasons. First, the presence of iris nodules is consistent with granulomatous iridocyclitis, which is most likely to occur with granulomatous diseases. The elevated lysozyme level, elevated erythrocyte sedimentation rate, and presence of hilar and diffuse lymphadenopathy are consistent with the diagnosis of sarcoidosis. Sarcoidosis is a systemic granulomatous disease of unknown cause that is associated with immune activation and disease activity and typically responds to treatment with immunosuppression. Although few findings are diagnostic of sarcoidosis, elevated levels of the biomarkers lysozyme and angiotensin-converting enzyme are helpful; an elevated lysozyme level is more specific. Lysozyme is an antimicrobial enzyme that is produced by monocytes and macrophages, and the level is elevated in patients with sarcoidosis because macrophages are activated.[2]

TUBERCULOSIS

Although I think that sarcoidosis is the most likely diagnosis in this patient, the recent positive interferon-γ release assay for *M. tuberculosis* gives me pause, because the treatment for sarcoidosis is immunosuppression. Tuberculosis is the second most likely diagnosis in this patient, given her epidemiologic risk and positive interferon-γ release assay. In addition, tuberculosis is known to mimic other diseases.

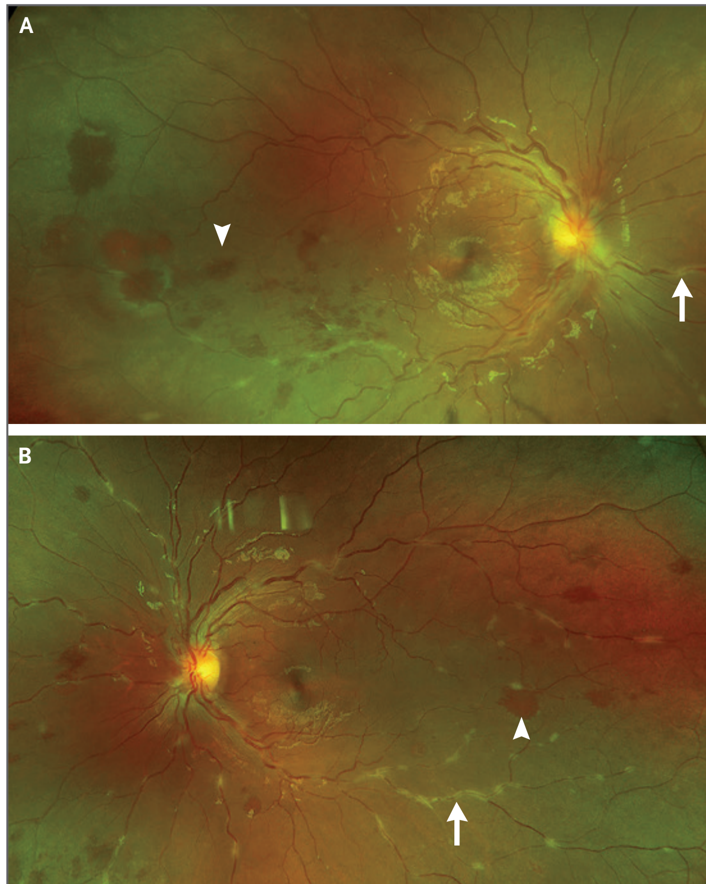

Figure 2. Fundus Photographs.
Fundus photographs of the right eye and the left eye (Panels A and B, respectively) show perivascular exudates (arrows) and hemorrhages (arrowheads).

OTHER GRANULOMATOUS DISEASES

GPA is a systemic disease that can have manifestations similar to those of sarcoidosis. However, this diagnosis is unlikely in this patient because she had a negative test for antineutrophil cytoplasmic antibodies.

The Blau syndrome (also known as juvenile systemic granulomatosis) is caused by a gain-of-function mutation in *NOD2*, the gene encoding nucleotide-binding oligomerization domain containing 2, which is a type of pattern-recognition receptor. This defect leads to immune activation and the early onset (at <5 years of age) of a triad of symptoms: arthritis, uveitis, and dermatitis. The

Blau syndrome is unlikely in this patient because she first presented with symptoms at 11 years of age.

Chronic granulomatous disease is an inherited immunodeficiency that is caused by defects in phagocyte NADPH oxidase, which is responsible for the phagocytic respiratory burst. Like patients with many other immunodeficiency syndromes, patients with chronic granulomatous disease have an increased susceptibility to infection with specific pathogens and have autoimmune disease that is difficult to control. Some patients with immunodeficiency present without a history of severe infections. Nevertheless, chronic granulomatous disease is unlikely in this patient, because the condition is rare and she had been completely healthy up until this point.

Several other, more common autoimmune diseases can be associated with granuloma formation. Crohn's disease is one of these conditions, but it is unlikely in this patient because she did not have fever, weight loss, or gastrointestinal symptoms. Similarly, Behçet's disease can cause granulomatous uveitis, but oral and genital lesions are most likely to develop. Finally, Susac's syndrome can cause retinal vasculopathy, but sensorineural hearing loss and encephalopathy usually occur, and these findings were not present in this patient.

To establish the diagnosis of sarcoidosis, a tissue diagnosis is usually required. In this case, a lymph-node biopsy would most likely reveal noncaseating granulomas, a feature consistent with the diagnosis of sarcoidosis.

CLINICAL IMPRESSION

Dr. Lucia Sobrin: This patient had bilateral granulomatous panuveitis with retinal vasculitis in the context of diffuse lymphadenopathy involving the chest and abdomen and a recent positive interferon-γ release assay for *M. tuberculosis*. The two entities of greatest concern are indeed sarcoidosis and tuberculosis. An acid-fast bacilli smear and culture of sputum that had been obtained during her previous hospitalization were negative. A tissue diagnosis would be the most definitive way to differentiate between these two diseases. However, a lymph-node biopsy was not pursued, because the location of the involved lymph nodes in the chest meant that thoracoscopic surgery would need to be performed rather than bronchoscopy, and the care team did not think that the risks associated with surgery were warranted.

It is not uncommon for ophthalmologists to have patients with uveitis in whom the diagnosis remains unclear after results of serologic testing and imaging are received. An approach that is sometimes helpful in determining an infectious cause of uveitis is sampling of ocular fluid for nucleic acid amplification testing. However, such testing of ocular fluid is not sensitive for the diagnosis of tuberculosis,[3] and ophthalmologists are often left with the option of providing empirical treatment for possible tuberculosis uveitis and following the patient closely for a response to treatment. In this patient, the four-drug regimen was continued as empirical treatment for possible tuberculosis. The course of prednisone was continued but then tapered gradually, and glucocorticoid-sparing therapy with mycophenolate mofetil was started eventually. Scatter laser treatment to the areas of retinal nonperfusion in both eyes was applied. Fortunately, the patient's uveitis became quiescent.

CLINICAL DIAGNOSIS

Sarcoidosis or tuberculosis.

DR. EZRA M. COHEN'S DIAGNOSIS

Sarcoidosis.

DISEASE COURSE

Dr. Diacovo: One year after the patient was evaluated in the rheumatology clinic of this hospital and 3 months after she completed a 9-month course of medication for possible tuberculosis, a "wobbly" gait developed while she was receiving therapy with mycophenolate mofetil (1500 mg per day). The walking difficulty worsened, such that the patient had to hold her mother's arm to walk. The patient's parents called the rheumatology clinic, and a telemedicine visit was arranged promptly. On examination during the video call, the patient was observed to have a bilateral Trendelenburg gait. She could not stand on one leg, stand on her toes, or squat. She was instructed to present to the emergency department of this hospital.

In the emergency department, the patient described aching pain in the lower back and both hips and weakness in the lower legs that had developed 10 days earlier. She reported no fever, vision changes, sensation changes, incontinence, recent illness, or trauma. Four weeks before this evaluation, she had received a seasonal influenza vaccination. On examination, she appeared well. There was profound weakness on eversion of the right ankle, dorsiflexion of both ankles, and flexion of the left knee and both hips. Strength and reflexes in the arms were normal, as was sensation in response to light touch, pain, temperature, and vibration. Results of finger-to-nose testing were normal; the Romberg sign and the Babinski reflex were absent. There was a wide-based gait, and the right foot dragged more than the left foot.

Dr. Figueiro Longo: MRI of the lumbar spine (Fig. 3A and 3B), performed before and after the administration of intravenous contrast material, revealed an enhancing, T2-hyperintense intramedullary lesion in the lower thoracic spinal cord and conus medullaris. There was also smooth thickening and enhancement of the cauda equina nerve roots. MRI of the head (Fig. 3C), performed before and after the administration of intravenous contrast material, revealed prominence of the lacrimal glands.

SECOND DIFFERENTIAL DIAGNOSIS

Dr. Danielle B. Pier: I participated in the care of this patient and am aware of the final diagnosis. This patient presented with weakness in the lower legs that was affecting ambulation. Strength was preserved elsewhere, so a cerebral or cervical lesion seemed unlikely. Sensation was spared, and reflexes were preserved initially. The Guillain–Barré syndrome was considered briefly, given her history of recent vaccination, although preserved reflexes are not consistent with this diagnosis. Spinal imaging then revealed an expansile lesion at the level of T12–L1 that extended into the cauda equina nerve roots, a finding suggestive of a myeloradiculopathy. The Guillain–Barré syndrome does not involve the spinal cord and thus was ruled out in this case. The findings on imaging were also not consistent with ischemia or hemorrhage. The most likely causes of this patient's lower-leg weakness were infection, neoplasm, and autoimmune disease.

INFECTION

The patient had recently completed a 9-month course of antituberculosis medication. She had also received immunosuppressive therapy, which increased her risk of infection. Tuberculosis with spinal cord involvement commonly manifests as a myeloradiculopathy, although other possible manifestations include intramedullary or extramedullary tuberculomas, spinal abscesses, acute myelitis, longitudinal extensive transverse myelitis, syringomyelia, cauda equina syndrome, and spinal cord infarction.[4] This patient had lived in the northeastern United States for the previous 6 years, so neuroborreliosis was another consideration, but the absence of cranial nerve involvement would be unusual and a screening test for Lyme disease was negative. Other infections — such as syphilis and

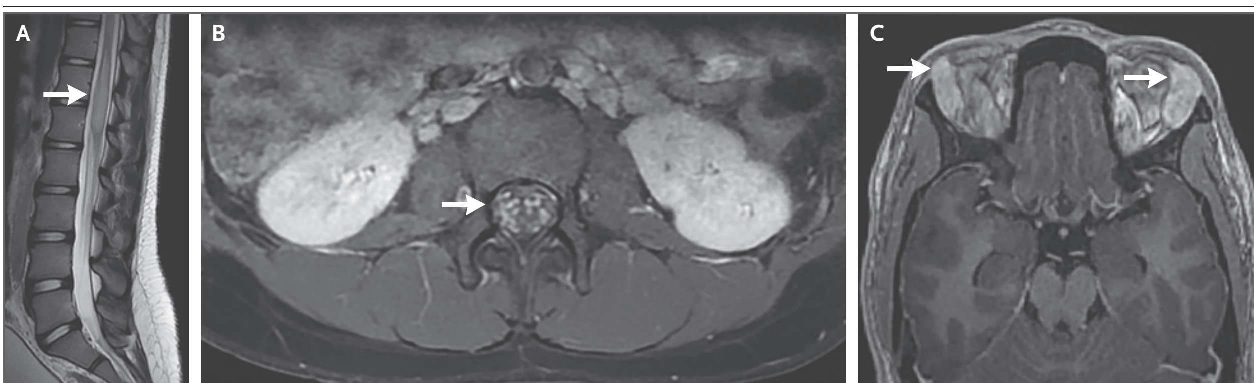

Figure 3. MRI of the Lumbar Spine and Head.

A sagittal T2-weighted image of the lumbar spine (Panel A) shows a hyperintense intramedullary, slightly expansive lesion in the spinal cord at the level of T12–L1 (arrow). An axial T1-weighted fat-suppressed contrast-enhanced image of the lumbar spine (Panel B) shows smooth thickening and enhancement of the cauda equina nerve roots (arrow). An axial T1-weighted fat-suppressed contrast-enhanced image of the head (Panel C) shows prominence of the lacrimal glands (arrows).

cytomegalovirus, varicella–zoster virus, herpes simplex virus, West Nile virus, HIV, and JC virus infections — can manifest as a myeloradiculopathy in rare cases, such as cases in which the patient is severely immunocompromised. Fungal and parasitic infections can involve the spinal cord and nerve roots, although the development of brain lesions would be more common.

Neoplasm

A primary spinal cord tumor was unlikely in this patient because the symptoms had developed just 10 days earlier. Perhaps a particularly aggressive tumor, such as a diffuse fibrillary astrocytoma, could progress this quickly, but symptoms of primary spinal cord tumors most often develop over a period of months to years. Also, most primary spinal cord tumors, including low-grade gliomas and ependymomas, have discrete margins.[5] Germ-cell tumors were another consideration in this patient, although they are more likely to occur in the suprasellar region with subsequent spread to the spinal cord. Of note, germ-cell tumors can cause granulomatous inflammation that can mimic other granulomatous diseases.[6] Many malignant spinal cord lesions originate from solid organ tumors, non-Hodgkin's lymphoma, or leukemia. Primary central nervous system (CNS) lymphoma typically develops in the brain, but CNS lymphoma with isolated spinal cord involvement has been reported.[7]

Autoimmune Disease

Neuroimmunologic conditions most often involve the CNS, but peripheral nervous system (PNS) involvement has been reported with some disorders, including multiple sclerosis,[8] neuromyelitis optica,[9] disorders associated with antibodies to myelin oligodendrocyte glycoprotein (MOG),[10] disorders associated with antibodies to glial fibrillary acidic protein (GFAP),[11] and transverse myelitis.[12] Because this patient had recently received a clinical diagnosis of uveitis, a systemic condition affecting the spinal cord was the most likely cause of her leg weakness. Such conditions include sarcoidosis, systemic lupus erythematosus, Sjögren's syndrome, common variable immunodeficiency,[13] and systemic vasculitides.[14]

Sarcoidosis was the most likely diagnosis, given her recent episode of uveitis. Like tuberculosis, sarcoidosis can affect any part of the neurologic system and thus can cause almost any neurologic sign or symptom, including meningeal inflammation (aseptic meningitis), cerebral or cerebellar lesions, neuroendocrine dysfunction, optic chiasm compression, brainstem lesions, cranial neuropathies, spinal cord lesions, conus medullaris or cauda equina syndrome, radiculopathy, polyneuropathy, myopathy, and cerebrovascular accidents related to vascular inflammation.[15] Although this patient's blood level of angiotensin-converting enzyme was normal, no serum or cerebrospinal fluid test can either confirm or rule out the diagnosis of sarcoidosis.[16] On the basis of the 2018 proposed diagnostic criteria for CNS and PNS neurosarcoidosis,[17] it became apparent that this patient should undergo a tissue biopsy. MRI of the head had shown prominent lacrimal glands, so we opted for a lacrimal gland biopsy because it was the least invasive approach to establishing the diagnosis of sarcoidosis.

Pathological Discussion

Dr. Anna M. Stagner: A lacrimal gland biopsy was performed. Hematoxylin and eosin staining of sections of the tissue specimen showed overall preservation of the gland, with normal architecture. There were intact lobules of acini that were rich in zymogen granules and unremarkable lacrimal ducts. However, within the parenchyma, there were abnormal discrete pale nodules (Fig. 4A). At higher magnification, these nodules were shown to be well-defined granulomas (Fig. 4B), which are sometimes called "naked granulomas" because they are composed predominantly of histiocytes, without many other admixed inflammatory cells such as neutrophils. The composite epithelioid histiocytes had indistinct cell borders and abundant pale-pink cytoplasm, and they formed multiple distinct syncytial aggregates with scattered multinucleated giant cells (Fig. 4C). Most of the granulomas were nonnecrotizing or noncaseating; rare granulomas had foci of early central necrosis (Fig. 4D).

A Gomori methenamine–silver stain for fungi was negative, as were acid-fast and Fite stains for mycobacteria. Because small foci of necrosis were detected — and at the time of biopsy, there was an unclear history regarding positive testing for tuberculosis — nucleic acid amplification testing for tuberculous and nontuberculous mycobacteria was performed, and it was also negative.

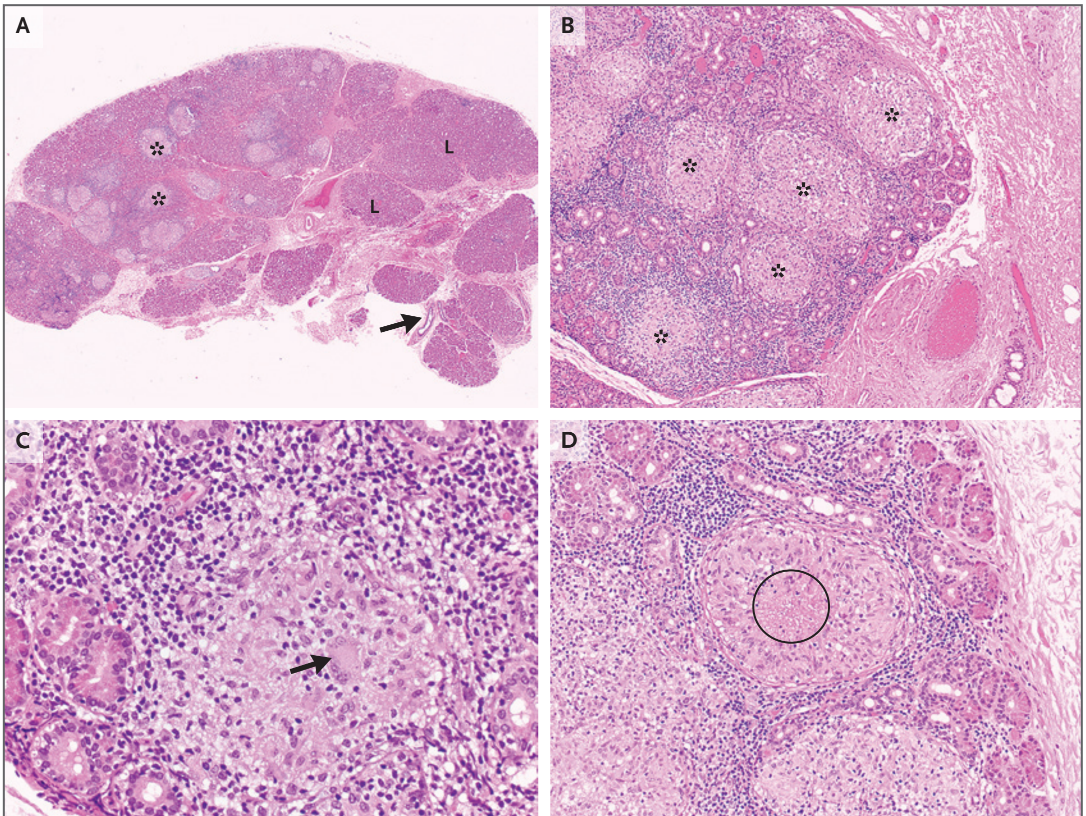

Figure 4. Lacrimal Gland–Biopsy Specimen.

Hematoxylin and eosin staining of the lacrimal gland–biopsy specimen (Panel A) shows discrete pale nodules (asterisks) within the otherwise normal lacrimal gland, with intact lobules of acini (L) and lacrimal ducts (arrow). At higher magnification (Panels B, C, and D), well-defined epithelioid granulomas (Panel B, asterisks) are present within the lacrimal gland parenchyma, surrounded by normal acini. The epithelioid histiocytes have abundant eosinophilic cytoplasm and are admixed with multinucleated giant cells (Panel C, arrow). Rare granulomas have foci of central necrosis (Panel D, circled).

In a patient with predominantly nonnecrotizing, well-defined epithelioid granulomas in the lacrimal gland, the most likely diagnosis is sarcoidosis.[18,19] Infectious processes, particularly fungal and mycobacterial infections, must be ruled out. GPA and rheumatoid nodules can be associated with extensive necrosis, including palisading granulomas, in which case well-formed epithelioid granulomas would not be present; GPA is also associated with vasculitis.[20] Finally, extraintestinal Crohn's disease may cause granulomatous orbital inflammation.[21] In this case, after considering the findings on histologic examination in the context of the clinical features, and after ruling out mycobacterial infection, the pathological diagnosis was sarcoidosis.

PATHOLOGICAL DIAGNOSIS

Sarcoidosis.

DISCUSSION OF MANAGEMENT

Dr. Daniel R. Brown: It is important to recognize that sarcoidosis is a variable disease.[22] For example, early-onset sarcoidosis manifests as the triad of granulomatous uveitis, arthritis, and rash and typically occurs in patients younger than 5 years of age; some patients with early-onset sarcoidosis are found to have mutations in *NOD2* and ultimately receive a diagnosis of the Blau syndrome. In addition, there are acute febrile presentations of sarcoidosis that are often self-limited, including the Löfgren syndrome, which is associated with erythema nodosum, hilar lymphadenopathy, and polyarthritis, as well as the Heerfordt syndrome, which consists of uveitis, parotid enlargement, and facial nerve paralysis. The manifestations of sarcoidosis in adolescents (such as this patient) are similar to those observed in adults, with only minor differences in the reported incidence of various symptoms. Therefore, the management approach depends on the patient's age at presentation, the organ systems involved, and whether the patient has one of these unique subtypes, all of which can inform the anticipated prognosis.

With this caveat, the general approach to treating sarcoidosis is to control immune-mediated damage from granuloma formation. For patients with asymptomatic disease and no organ dysfunction, treatment can be minimal. For patients with forms of disease that may compromise function of the eyes, heart, or CNS, more aggressive treatment is required. As first-line treatment, a course of moderate-to-high-dose glucocorticoids is typically provided, with tapering over a period of several months. If symptoms recur with tapering, glucocorticoid-sparing agents are added. Possible glucocorticoid-sparing agents include methotrexate, azathioprine, hydroxychloroquine, and mycophenolate mofetil, with the choice of medication depending on the severity of disease.

For uveitis, this patient was treated with a 5-day course of methylprednisolone (1 g per day) followed by a tapering course of prednisone and then mycophenolate mofetil. Unfortunately, neurologic symptoms developed while she was receiving treatment. Retrospective studies investigating the use of mycophenolate mofetil in the treatment of neurosarcoidosis have shown mixed results. An early small retrospective case series showed remission in the majority of patients,[23] whereas a later retrospective study comparing methotrexate with mycophenolate mofetil showed a greater relapse rate among patients who had been treated with mycophenolate mofetil.[24]

Helper T cells (CD3+CD4+ T lymphocytes) are crucial for granuloma formation, and the tumor necrosis factor α (TNF-α) that is produced by tissue macrophages amplifies the granuloma reaction. This process forms the rationale for trials investigating the use of TNF-α blockers in the treatment of sarcoidosis. An early placebo-controlled trial involving patients with pulmonary sarcoidosis showed a significant improvement in forced vital capacity, but the effect was small and not considered to be clinically significant.[25] Interest in the use of TNF-α blockers remains, and retrospective studies investigating the use of infliximab in the treatment of neurosarcoidosis have shown a positive effect.[26,27]

This patient was treated with another 5-day course of methylprednisolone (1 g per day) followed by a course of prednisone that was tapered over a period of 6 months. After 3 months of therapy, all symptoms had resolved, and infliximab was added to facilitate the tapering of the prednisone course. Unfortunately, the patient stopped taking all medications, and uveitis recurred within 5 months after her last infliximab infusion. Medications have since been restarted.

FINAL DIAGNOSIS

Sarcoidosis.

This case was presented at the Harvard Medical School postgraduate course "Pediatric Primary Care," directed by Peter T. Greenspan, M.D., John Patrick T. Co, M.D., M.P.H., Ronni L. Goldsmith, M.D., Janice A. Lowe, M.D., and Benjamin A. Nelson, M.D.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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